

Supplemental Material for “Projective Outcome Geometry and Born-Like Dipoles in Unitary Many-Body Detectors”

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This Supplemental Material fixes the mathematical conventions, proves the outcome-antipode and Haar results, reconciles the forward construction with the production calculation, and records the numerical methods and provenance. It is single-column throughout. The term “collapse” is operational: the construction identifies product boundary-compatible states, not a discontinuous global dynamics or a physical selection law.

FORWARD POLE CONSTRUCTION

Let $U : \mathbb{C}^2 \otimes \mathbb{C}^d \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^d$ be unitary, with qubit index first and detector basis fixed,

$$U = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (\text{S1})$$

Write the input qubit ray homogeneously as $q = [\alpha : \beta] \in \mathbb{CP}^1$ and let $\eta \neq 0$ be a detector vector. Direct block multiplication gives

$$U[(\alpha|0\rangle + \beta|1\rangle) \otimes \eta] = |0\rangle \otimes (\alpha A + \beta B)\eta + |1\rangle \otimes (\alpha C + \beta D)\eta. \quad (\text{S2})$$

Thus output pole 0 requires

$$L_0(\alpha, \beta)\eta = (\alpha C + \beta D)\eta = 0, \quad (\text{S3})$$

whereas pole 1 requires $L_1(\alpha, \beta)\eta = (\alpha A + \beta B)\eta = 0$. Unitarity ensures that the surviving component in Eq. (S2) is nonzero. The affine coordinate $z = \beta/\alpha = e^{i\phi} \tan(\theta/2)$ gives $C + zD$ and $A + zB$; $[0 : 1]$, or $z = \infty$, remains part of the projective problem.

EXACT ANTIPODAL OUTCOME DUALITY

Choose a representative with $|\alpha|^2 + |\beta|^2 = 1$ and define its orthogonal coordinate

$$q^\perp = (-\bar{\beta}, \bar{\alpha})^T, \quad R_q = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \in SU(2). \quad (\text{S4})$$

The lower-left block of $V = U(R_q \otimes I_d)$ is $L_0(q)$ and the upper-right block is $L_1(q^\perp)$. Jacobi’s complementary-minor identity applied to the unitary V yields

$$\det L_0(\alpha, \beta) = (-1)^d \det U \overline{\det L_1(-\bar{\beta}, \bar{\alpha})}. \quad (\text{S5})$$

Therefore q is an outcome-0 root if and only if $q^\perp$ is an outcome-1 root. Algebraic multiplicity is preserved. In stereographic coordinates the map is

$$z \mapsto z_\perp = -\frac{1}{\bar{z}}, \quad (\text{S6})$$

with $0 \leftrightarrow \infty$. The two pencils are regular or singular together.

For identically processed empirical measures,

$$\rho_1(\Omega) = \rho_0(-\Omega), \quad a(-\Omega) = -a(\Omega). \quad (\text{S7})$$

Hence a spherical-harmonic expansion of the exact paired asymmetry contains only odd ℓ . A projective Born rule along $\hat{\mathbf{n}}$ is the $\ell = 1$ field $a_B = \hat{\mathbf{n}} \cdot \mathbf{r}$. Equivalently, with $s_{\text{even}}(\Omega) = \rho_0(\Omega) + \rho_0(-\Omega)$,

$$\rho_0(\Omega) = \frac{s_{\text{even}}(\Omega)}{2} [1 + \hat{\mathbf{n}} \cdot \mathbf{r}(\Omega)]. \quad (\text{S8})$$

The total density may be highly nonuniform. For Born behavior, its normalized odd part must be a unit dipole. Along z , Eq. (S8) is equivalent to $\rho_0(-\Omega)/\rho_0(\Omega) = \tan^2(\theta/2)$.

REGULAR ROOTS, INFINITY, AND SINGULAR PENCILS

For a regular pencil, $p_0(\alpha, \beta) = \det(\alpha C + \beta D)$ is a nonzero homogeneous polynomial of degree d . It has exactly d roots on $\mathbb{CP}^1$, counted with algebraic multiplicity. This does not imply d distinct Bloch points. At a root q_j with nullity g_j , compatible detector rays form $\mathbb{CP}^{g_j-1}$; algebraic multiplicity and nullity need not agree.

In the generalized-eigenvalue convention $bC\eta = aD\eta$, for which the forward coordinate is $z = -a/b$, a normalized residual should be evaluated without affine division:

$$r = \frac{\|bC\eta - aD\eta\|}{(|b|\|C\| + |a|\|D\|)\|\eta\|}. \quad (\text{S9})$$

A zero denominator coordinate is a root at infinity rather than a numerical failure. QZ/generalized Schur methods avoid explicit inversion and preserve this structure [1].

If $p_0 \equiv 0$, the finite-root statement fails and the Kronecker structure is required. Qubit-qubit SWAP is the clean example:

$$U_{\text{SWAP}}(|q\rangle_Q |0\rangle_D) = |0\rangle_Q |q\rangle_D, \quad (\text{S10})$$

so every input qubit reaches pole 0. Conversely, $U = I$ is regular but maximally degenerate: each outcome has one Bloch pole with algebraic multiplicity d and a d -dimensional detector nullspace.

The full pole-preimage set $\mathcal{L}_k = U^\dagger(|k\rangle \otimes \mathcal{H}_D)$ is a d -dimensional linear subspace. The special product cone is $\mathcal{S}_k = \mathcal{L}_k \cap \Sigma$, where Σ is the Segre product cone. If a regular pencil has two distinct qubit roots q_1, q_2 , compatible detector vectors cannot be proportional: otherwise both detector blocks would share a kernel and the pencil would be singular. The sum $q_1 \otimes \eta_1 + q_2 \otimes \eta_2$ therefore has Schmidt rank two. Generic nonadditivity belongs to the product intersection $\mathcal{S}_k$, not to the linear preimage space $\mathcal{L}_k$.

HAAR ROOTS ARE EXACTLY THE SPHERICAL ENSEMBLE

For the forward outcome-0 pencil, the adjoint block row

$$Q = \begin{pmatrix} C^\dagger \\ D^\dagger \end{pmatrix} \quad (\text{S11})$$

is a Haar Stiefel frame. If G_0, G_1 are independent complex $d \times d$ Ginibre matrices, then

$$Q \stackrel{d}{=} \begin{pmatrix} G_0 \\ G_1 \end{pmatrix} (G^\dagger G)^{-1/2}. \quad (\text{S12})$$

Taking adjoints shows that $C + zD$ contains a common invertible left factor. Its generalized roots are therefore exactly those of $G_0^\dagger + zG_1^\dagger$, the complex spherical ensemble [2, 3]. The joint density is proportional to

$$\prod_{i < j} |z_i - z_j|^2 \prod_{j=1}^d (1 + |z_j|^2)^{-(d+1)}, \quad (\text{S13})$$

and the one-point intensity is

$$\rho^{(1)}(z) = \frac{d}{\pi} (1 + |z|^2)^{-2} = \frac{d}{4\pi} \quad \text{per unit solid angle.} \quad (\text{S14})$$

Equation (S14) is exact at every d . It does not state that one finite realization is uniform or provide a concentration rate. The same Stiefel-factor cancellation applies to the production pencil introduced next.

There is also a direct covariance check. For $R \in SU(2)$, right multiplication $U \mapsto U(R \otimes I_d)$ sends the forward root set by the inverse projective rotation $q \mapsto R^{-1}q$. Haar right invariance therefore makes the root law $SU(2)$ covariant. Since $SU(2)$ acts transitively on the Bloch sphere, covariance plus the exact count d forces the uniform one-point intensity in Eq. (S14); it does not force a finite cloud to be uniform. Left multiplication gives the analogous covariance of the production-output roots.

PRODUCTION PENCIL AND TIME-DIRECTION BRIDGE

The post-cutoff campaigns did not explicitly form the forward blocks (C, D) and (A, B) . They fixed the input pole and solved

$$Cv = \lambda Av. \quad (\text{S15})$$

For a finite root,

$$U(|0\rangle \otimes v) = (|0\rangle + \lambda|1\rangle) \otimes Av. \quad (\text{S16})$$

Thus production enumerates separable outputs. Applying $U^\dagger$ to Eq. (S16) shows that the same qubit roots are exact forward pole-0 preimages for $U^\dagger = U(-t)$; Eq. (S5) supplies the antipodal outcome-1 root coordinates for that inverse propagator. The corresponding detector vectors for outcome 1 are not stored.

For the highlighted Hamiltonian, the matrix is real and symmetric in the stated computational basis. Hence $U(-t) = U(t)^*$ and production root multisets obey $\lambda_j(-t) = \lambda_j(t)^*$. Polar radii are invariant and azimuths reflect. The Letter displays production ϕ ; the same-time forward map is obtained by $\phi \mapsto -\phi$. This reflection leaves the z -axis Born target and every quoted polar statistic invariant.

The campaign code formed the dense relative matrix through a linear solve and stored its ordinary eigenvalues. It did not store homogeneous QZ pairs, block condition estimates, backward residuals, nullities, or a singular-pencil classification for the $N = 11$ –16 sequence. All stored roots in the included N16 arrays are finite, but that fact alone is not a conditioning audit. A homogeneous-QZ reproduction is therefore a submission-critical numerical check, not a completed validation.

HAMILTONIAN AND RUN PROVENANCE

The detector has periodic bonds and the measured qubit is site 0. With dimensionless Pauli operators and $\hbar = 1$,

$$\begin{aligned} H = & -h_{z0}Z_0 - h_z \sum_{i=1}^N Z_i - J \sum_{i=1}^N Z_i Z_{i+1} - J_\pm \sum_i (\sigma_i^+ \sigma_{i+1}^- + \text{H.c.}) \\ & - \frac{J_x}{\sqrt{N}} X_0 \sum_i X_i - \frac{J_y}{\sqrt{N}} Y_0 \sum_i Y_i, \end{aligned} \quad (\text{S17})$$

where $Z_{N+1} = Z_1$. The matched family uses $h_{z0} = h_z = 0.1$, $J = 1$, $J_\pm = J_y = 0$, and collective $J_x = 0.01$; the per-edge X coupling is $J_x/\sqrt{N}$. Equation (S17) follows the repository sign convention.

The Zeus completion records span July 14–15, 2026. The N16 run began at 2026-07-14T20:07:01Z and finished at 2026-07-15T14:15:06Z using Python 3.11.5, NumPy 2.2.6, SciPy 1.17.1, QuSpin 1.0.0, and QuSpin-Extensions 0.1.6. It stored $2^{16} = 65,536$ roots at each of $t = 10^3, 10^4, 10^5, 10^6$. No disorder or random seed enters this clean matched family.

The finite-size analysis table contains 1,208 completed spectra: 168 central-field, 300 detector-field-resonance, 240 fixed-exchange, and 500 exchange-plus-field rows. The declared descriptive gates are polar coverage ≥ 0.50 and $|c_2| \leq 0.25$. Exactly 24 rows pass, all from the matched family; they are the four times at each $N = 11, \dots, 16$. This is a deterministic parameter-grid observation, not a population frequency.

TABLE S1. Matched-field finite-size summary. Medians and maxima are over four deterministic saved times, not independent stochastic repeats.

N	roots/spectrum	median S_B	maximum S_B	median $ c_2 $
11	2,048	0.19155	0.31964	0.02830
12	4,096	0.41110	0.46636	0.02439
13	8,192	0.57147	0.61996	0.02520
14	16,384	0.71519	0.75282	0.01526
15	32,768	0.78839	0.80310	0.01904
16	65,536	0.80747	0.83950	0.01672

TABLE S2. Binning sensitivity of the N16, $t = 10^4$ full-sphere map. Finer grids expose finite-count fluctuations; no continuum extrapolation is attempted.

$N_\phi \times N_\mu$	empty	minimum count	ϵ_B	correlation	dipole-fit residual
24×12	0	34	0.09039	0.98477	0.05343
36×18	0	10	0.08909	0.97150	0.07024
48×24	0	4	0.11539	0.92730	0.10551
72×36	16	1	0.14764	0.85147	0.14206

FULL-SPHERE METHODOLOGY AND ROBUSTNESS

For a production root λ , the Bloch coordinates are

$$\mathbf{r}(\lambda) = \frac{(2 \operatorname{Re} \lambda, 2 \operatorname{Im} \lambda, 1 - |\lambda|^2)}{1 + |\lambda|^2}, \quad \phi = \arg \lambda, \quad \mu = \cos \theta = \frac{1 - |\lambda|^2}{1 + |\lambda|^2}. \quad (\text{S18})$$

The antipodal outcome coordinate is $(-\mu, \phi + \pi)$ modulo 2π . Equal-width bins in (ϕ, μ) have equal area. If n_0, n_1 are the two label counts, the displayed field is $a = (n_0 - n_1)/(n_0 + n_1)$ without smoothing. The count-weighted residual is

$$\epsilon_B = \left[\frac{\sum_b (n_{0b} + n_{1b})(a_b - \mu_b)^2}{\sum_b (n_{0b} + n_{1b})} \right]^{1/2}. \quad (\text{S19})$$

A free dipole is obtained by weighted least squares against (r_x, r_y, r_z) at bin centers.

For $N = 16, t = 10^4$, the primary 36×18 grid has no empty bins and at least 10 labelled roots per bin. It gives $\epsilon_B = 0.08909$, correlation 0.97150, fitted amplitude 1.06225, fitted axis $(-2.1 \times 10^{-7}, 8.9 \times 10^{-7}, 1)$, and post-fit residual 0.07024. An equal-area complex spherical-harmonic projection through $\ell = 7$ assigns 0.98180 of the odd power to $\ell = 1$ and 0.01820 to odd $\ell \geq 3$. Across the three fully occupied grids 24×12 , 36×18 , and 48×24 , the dipole fraction ranges from 0.952 to 0.985. The even-power fraction remains at roundoff ($\sim 10^{-31}$), providing a numerical check of exact antipodal parity rather than an independent physical result.

The raw file used for the Letter map is

```
work/zeus_single_pixel_atlas_scaling_2026-07-14/
hz0/N16/raw/N16/hz0.+0.1000/raw-t10000.npz,
```

with SHA-256

e467b5efa6a5b8b7e3e7333a06555007b629a26bbeac2b21cd3da27d88313d59. The post-processing script verifies this digest before rendering.

INTERPRETIVE BOUNDARY AND SUBMISSION-CRITICAL CHECKS

Established exactly are the forward matrix-pencil reduction, antipodal outcome duality, regular projective root count, generic nonadditivity of the product intersection, the inverse-time production bridge, and the Haar spherical-ensemble law. Established numerically within the stated finite datasets are the matched-ring size trend, the full-sphere binned diagnostics, and the control separation in Fig. S1.

Heavy-tail and Born-score evidence are distinct gates

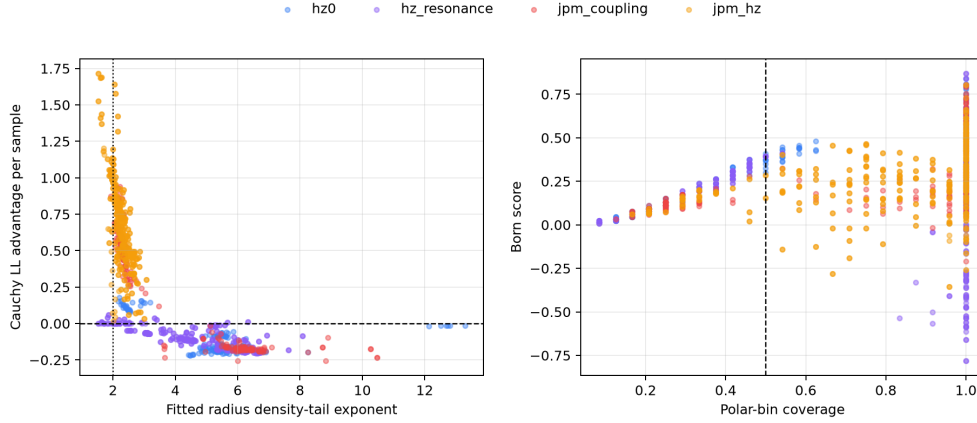


FIG. S1. **Post-cutoff control atlas.** Across 1,208 spectra, radial heavy-tail evidence, polar coverage, and the polar Born score are distinct diagnostics. Resonance and exchange families can show high polar scores while remaining nearly meridional. This descriptive figure is not a causal classifier and supplies no error bars.

Not established are an operational probability measure over roots and detector rays; a common apparatus-ready macrostate; stable, distinguishable, redundant detector records; homogeneous-QZ residual and conditioning validation of the production-scale roots; controlled $N \rightarrow \infty$, time-window, or resolution convergence; or a Hamiltonian-level mechanism explaining the matched condition. In particular, algebraic multiplicity is only a provisional counting convention. Spectral-statistics and chaos claims are excluded because the relevant spectra have not received the symmetry-sector and ensemble controls required for that inference [4].

The actionable missing calculations are maintained in `RESULTS_NEEDED.md`. The two most consequential are:

1. **[RESULT NEEDED: derive a normalized preparation/selection measure over qubit roots, detector null rays, degeneracies, and readout times without fitting the Born target];**
2. **[RESULT NEEDED: demonstrate that the associated detector factors form stable, distinguishable, preferably redundant record sectors over a nonzero time window].**

Without these results, the paper establishes measurement-compatible unitary geometry and Born-like root counts, not a completed ontology of definite outcomes.

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